# 5.2 Properties of Rational Exponents and Radicals

Name _____ Period ____ Date _____

Algebra II 5.2 · Student sheet

**I can…**

- simplify expressions with rational exponents.
- write radical expressions in simplest form.
- add, subtract, multiply, and divide radical expressions.

## Part 1 · True or false?

Use a calculator to test each statement. Mark it true or false. For each true one, name the property.

| Statement | True or false? | Property |
|---|---|---|
| 2^(1/2) · 2^(3/2) = $2^2$ | | |
| (5^(1/3))$^3$ = 5 | | |
| √2 · √8 = √16 | | |
| √2 + √3 = √5 | | |
| $\sqrt[3]{54} = 3\sqrt[3]{2}$ | | |
| √9 / √4 = √(9/4) | | |

Roots can be multiplied and divided, but they can't be _____

## Part 2 · Notes: fill in the blanks

$a^m \cdot a^n$ = _____ $(a^m)^n$ = _____ $a^m/a^n$ = _____

$\sqrt[n]{(a \cdot b)}$ = _____ $\sqrt[n]{(a/b)}$ = _____

A radical is in **simplest form** when the radicand has no perfect nth-power _____, there is no _____ in the radicand, and there is no radical in the _____.

The **conjugate** of $a + b\sqrt{c}$ is _____.

**Like radicals** have the same _____ and the same _____.

## Part 3 · Examples (work them with the class)

### Example 1

Simplify each expression.

a. $5^{(\frac{1}{3})} \cdot 5^{(\frac{5}{3})}$ b. $4^{\frac{\frac{5}{2}}{4}}{}^{(\frac{1}{2})}$ c. $(3^{(\frac{2}{3})})^3$

**Example 2**

Simplify each expression.

a. $\sqrt[3]{16} \cdot \sqrt[3]{4}$ b. $\sqrt[4]{80}/\sqrt[4]{5}$

**Example 3**

Write each expression in simplest form.

a. $\sqrt[5]{64}$ b. $\sqrt[4]{(\frac{5}{4})}$

**Example 4**

Write $\frac{1}{5+\sqrt{3}}$ in simplest form.

**Example 5**

Simplify each expression.

a. $7 \cdot \sqrt[5]{12} - \sqrt[5]{12}$ b. $\sqrt[3]{54} + \sqrt[3]{2}$

**Example 6**

Simplify each expression. Assume all variables are positive.

a. $\sqrt[3]{(64y^6)}$ b. $\sqrt[4]{(\frac{x^4}{y^8})}$

## Part 4 · Partner practice

### Partner 1

Write $\sqrt[3]{250}$ in simplest form.

### Partner 2

Write $\dfrac{3}{\sqrt{7}-2}$ in simplest form.